

```
1 # Create diabetes dataset
2 diabetes <- data.frame(
3   Age = c(21,22,23,24,25,26,27,28,29,30,
4           31,32,33,34,35,36,37,38,39,40,
5           41,42,43,44,45,46,47,48,49,50),
6   BloodPressure = c(65,70,68,72,75,69,74,76,73,78,
7                     80,77,79,81,82,78,84,83,85,86,
8                     87,88,84,89,90,91,92,88,93,94)
9 )
10
11 # Linear Regression
12 Relation <- lm(BloodPressure ~ Age, data = diabetes)
13
14 # Plot
15 plot(diabetes$Age,
16      diabetes$BloodPressure,
17      col = "green",
18      main = "Linear Regression Analysis",
19      xlab = "Age",
20      ylab = "BloodPressure")
21
22 abline(Relation, col = "red", lwd = 2)
23
24 # Model Summary
25 summary(Relation)
```

25:18 (Top Level)

R Script

Console Terminal Background Jobs

```
R - R 4.6.0 - ~/
> source("C:/Users/hoshe/Downloads/R Programming/multiple regression.R")
```

```
Call:
lm(formula = Age ~ BloodPressure + Glucose, data = Input)
```

```
Coefficients:
(Intercept) BloodPressure      Glucose
   -73.1835      0.9951      0.1639
```

```
(Intercept)
   -73.18355
BloodPressure
    0.9950987
Glucose
    0.1639295
```

```
Age = -73.18355 + 0.9950987 * BloodPressure + 0.1639295 * Glucose
> source("C:/Users/hoshe/Downloads/R Programming/Linear Regression.R")
>
```

Data

diabetes	30 obs. of 2 variables
diabetest1	10 obs. of 2 variables
Input	10 obs. of 3 variables
Model	List of 12
Relation	List of 12
result	List of 9

Values

A	Named num -73.2
tbl	'table' int [1:3, 1:2] 2 2 1 1 1 3
x	num [1:1000] -0.767 -0.816 -0.142 -0.278 0.436...

Linear Regression Analysis

